Predicting Marriage Compatibility Using TabTransformer

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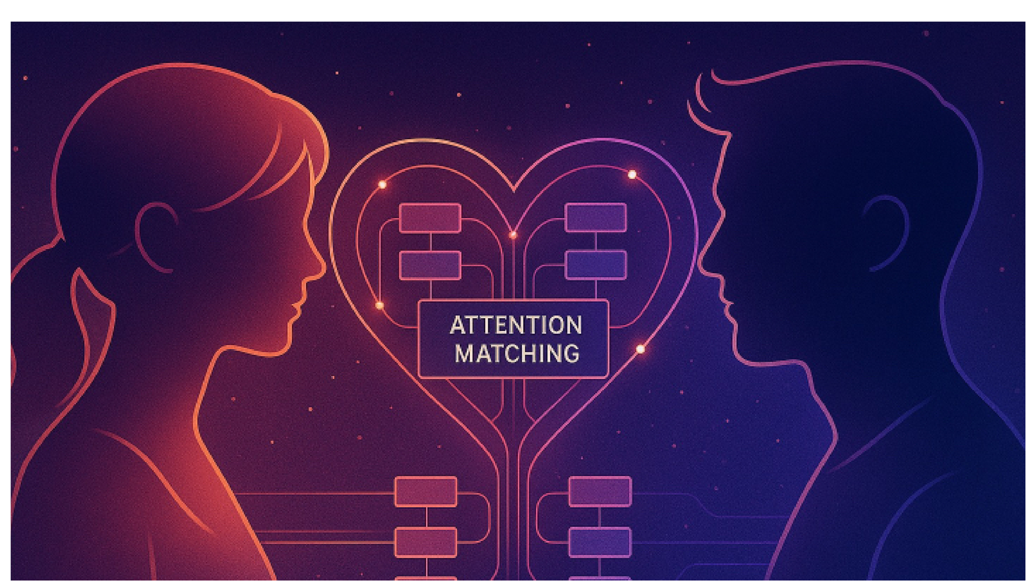
INTRODUCTION

- This study analyzes how **demographic traits** influence partner selection by predicting ideal match attributes from individual profiles. Using the **TabTransformer architecture** on **NYT wedding data**, we model relationships between features like age, education, and occupation through multi-label classification.
- Marriage patterns reveal societal structures and biases. By applying deep learning to partner prediction, we both **extend sociological research** and **enable practical applications** —from bias auditing to improving recommendation systems—while advancing Transformer methods for social data.



DATA PREPROCESS

1. Collection (NYT API 2012–2023)
   - Identified weddings via temporal rules:
     - Use NYT Article Search API and use both keywords like "wedding", "vow" and metadata subsection as "Fashion & Style"
2. Standardization Pipeline
   - **GPT Batch API Extraction** :
     - Extract core vars (age, school, job)
     - Normalized occupations (6 levels, 25 fields)
   - **School Categorization** (keyword search)
     - Ivy League / Top 50 Private / Top 50 Liberal Arts / Top 30 Public / Others
   - Age Binning:
     - Grouped into 5-year intervals: e.g., 20–24, 25–29

MODEL ARCHITECTURE

Our **ImprovedTabTransformer** is a neural architecture designed for tabular data, combining feature embeddings with transformer-based processing. Each categorical feature is first embedded into a shared space with LayerNorm for stable training. A learnable [CLS] token is added to aggregate global information.
The core consists of a 4-layer transformer encoder with 8 attention heads and 256-dimensional feedforward networks, enabling the model to learn complex feature interactions. For prediction, the [CLS] token's output is passed through task-specific heads—each a two-layer MLP with GELU activation and LayerNorm. This design efficiently captures both **feature-level patterns** and **global dependencies** while maintaining scalability.

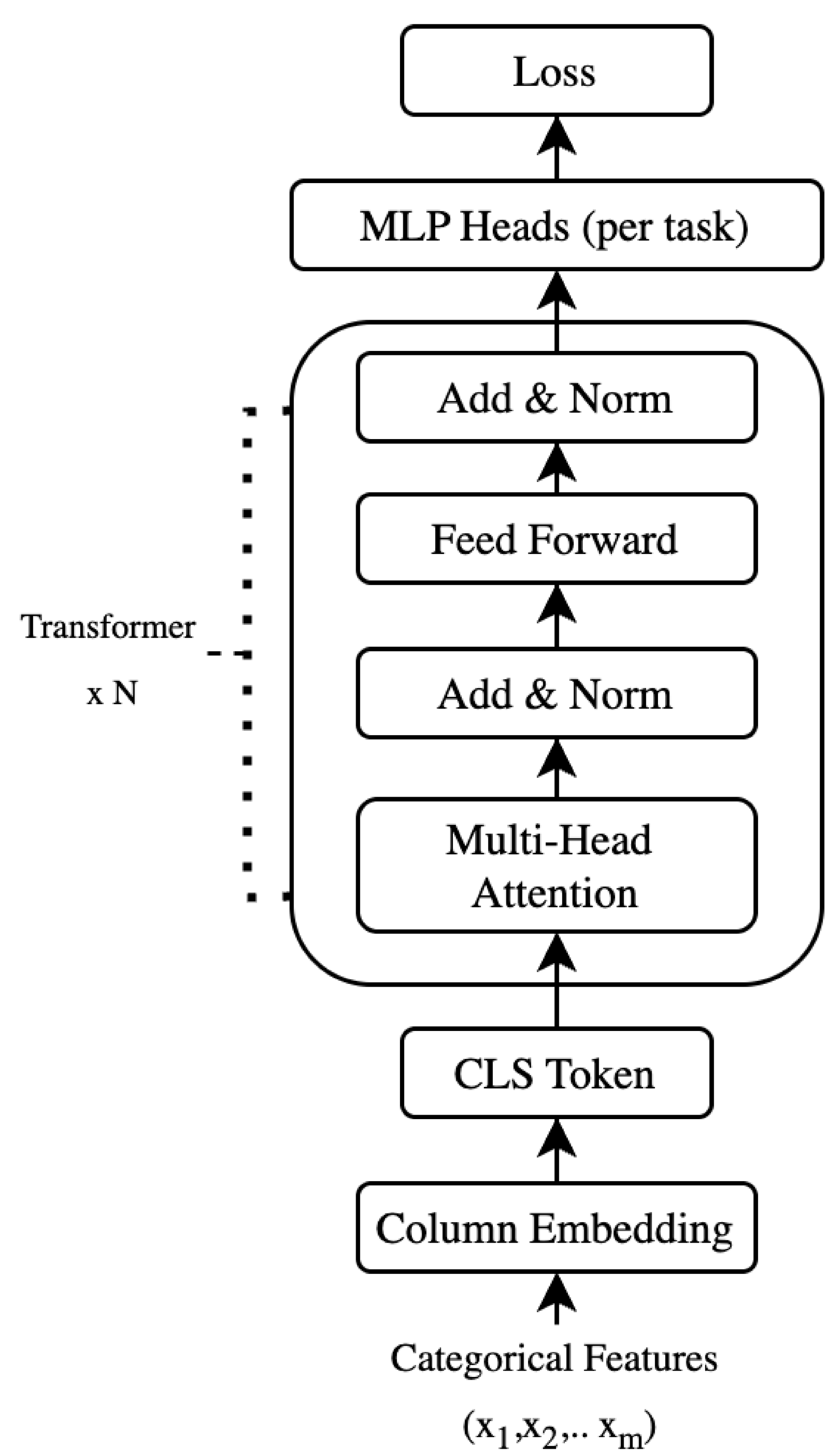


Figure1. Model Architecture.

RESULTS

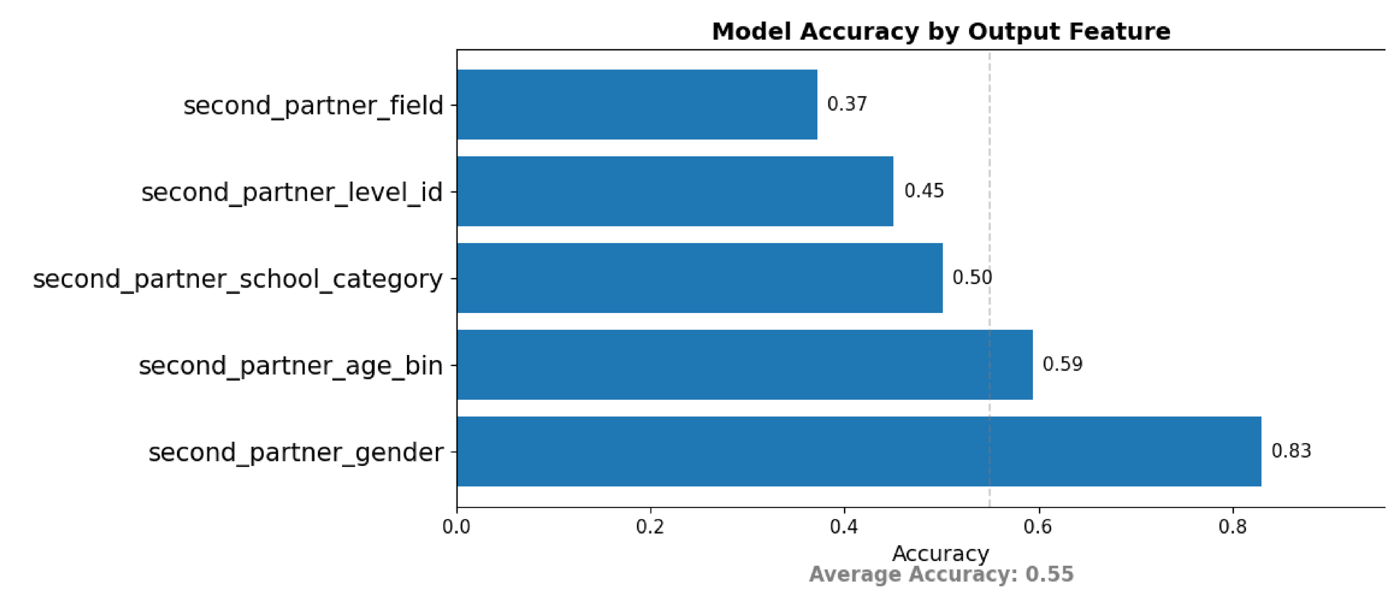


Figure 2. Model accuracy across five attribute prediction tasks.

- We evaluate our model's ability to predict structured partner attributes, including gender, age group, school category, education level, and field of study.
- As shown in the bar chart, the model achieves the highest accuracy on gender prediction (83%) and performs reasonably well on age group (59%). Performance on field of work is notably lower (37%), reflecting the complexity and variability of this attribute.
- The **average accuracy** across all five tasks is **55%**, demonstrating moderate model capability while leaving room for future improvement.

Attribute	Input Partner → Predicted Partner
Gender	Female → Male
Age Group	25–29 → 25–29
School Category	Ivy League → Ivy League
Education Level	S4 → S4
Field of Study	Computer & IT → Business & Finance

Table 1. Example Partner Prediction

- To qualitatively illustrate the model's behavior, we present a sample prediction case (right).
- Given a female partner aged 25–29 from an Ivy League background in IT, the model predicts a male partner of similar age and education level, but with a different professional background (Business & Finance).
- This example shows how the model c**aptures some plausible social pairing patterns** while allowing for **diverse outcomes** .

- While promising, the model's performance is affected by data limitations such as **class imbalance** and **societal selection bias**, as discussed in the Bias section.

DISCUSSION

- Our model performs best on highly structured and well-represented features, while struggling with more open-ended or imbalanced fields like occupation.
- This dataset is highly selective and not fully representative of the general population.
  - **Self-selection bias** : Only couples who choose to submit to the NYT (and are aware of the opportunity) are considered.
  - **Editorial bias** : The NYT editorial team curates which announcements are published, often favoring individuals with elite education, high-status professions, and East Coast connections.
- As a result, the dataset over-represents elite professions, Ivy+ schools, and white, cisgender individuals.

- Future work could incorporate more diverse datasets, use fairness-aware objectives, and explore multimodal inputs like text or images from wedding announcements. Such extensions would help generalize the model and reduce social reinforcement bias.

ACKNOWLEDGEMENTS

We analyze 9,160 ***New York Times*** wedding announcements (2013–2023 inclusively), collected via the **NYT API** and processed using **GPT-4** and rule-based methods to extract demographics (gender, age ranges, occupation, education). The dataset, annotated by **Dr. Zhenchao Qian** and **Dr. Guixing Wei** (Brown University), includes 60 structured variables, with methods adapted from [1].
[1] *Large-Scale Analysis of Marriage Dynamics in Social Media*(arXiv:2012.06678).